

Fig. 3: Author's prototype

D pins (D0, D1 and D2). Collector voltage is varied from 0V to 2.5V in eight steps to the digital-to-analogue converter from port B.

Output voltage from the digital-to-analogue converter is subsequently amplified about four times using LM358 and given to the base of the power transistor. Emitter of power transistor SL100 is connected to the collector of the transistor whose voltage is being varied.

Collector voltage is given to channel 0 of the built-in analogue-to-digital converter after dividing the voltage exactly by half using a potential divider arrangement to meet the voltage limitation of the microcontroller. Assuming that collector current is equal to emitter current, output is taken across the one-ohm resistance, and drop across one-ohm resistance is amplified about 50 times and given to channel 1 of the built-in analogue-to-digital converter.

The author's prototype is shown in Fig. 3.

Construction

An actual-size, single-side PCB layout for the USB interface is shown in Fig. 4 and its component layout in Fig. 5. After assembling the circuit on the PCB, enclose it in a suitable box.

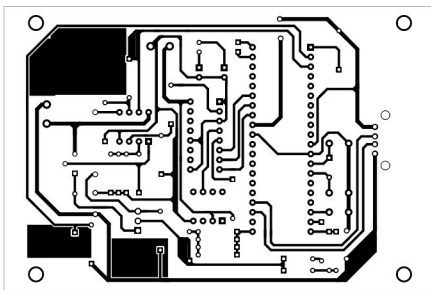


Fig. 4: Actual-size PCB layout of the USB interface using Python software

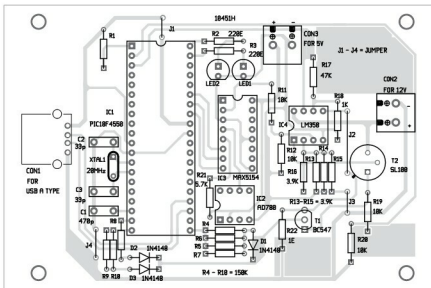


Fig. 5: Component layout of the PCB

Firmware

All USB devices handle a standard set of requests, described in the USB specification. It is implemented using pic18_usb header file provided in CCS C compiler. The compiler provides the basic framework for enumeration. During enumeration, the host requests data structures called descriptors from the device. These



Fig. 6: Hardware Wizard screen